Diabetes Regression Report - Train/Test Split

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Linear Regression evaluated on a standard train/test split.

MODEL OVERVIEW

Name	LinearRegression
Type	Regression
Sklearn	1.8.0
Parameter Count	13

DATASET OVERVIEW

Features	10
Total Observations	442
CV Folds	None
Train	331 (74.9%)
Test	111 (25.1%)

HYPERPARAMETERS

memory	None	transform_input	None
verbose	False	scaler	StandardScaler
model	LinearRegression	scaler__copy	True
scaler__with_mean	True	scaler__with_std	True

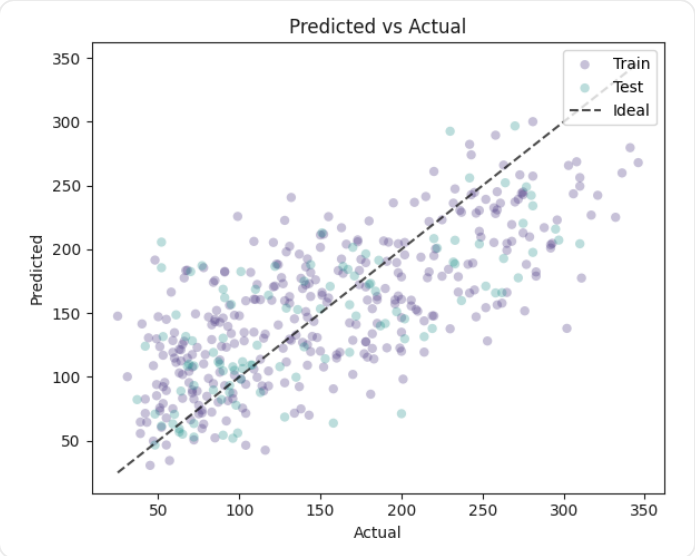
model__copy_X	True	model__fit_intercept	True
model__n_jobs	None	model__positive	False
model__tol	1e-06		

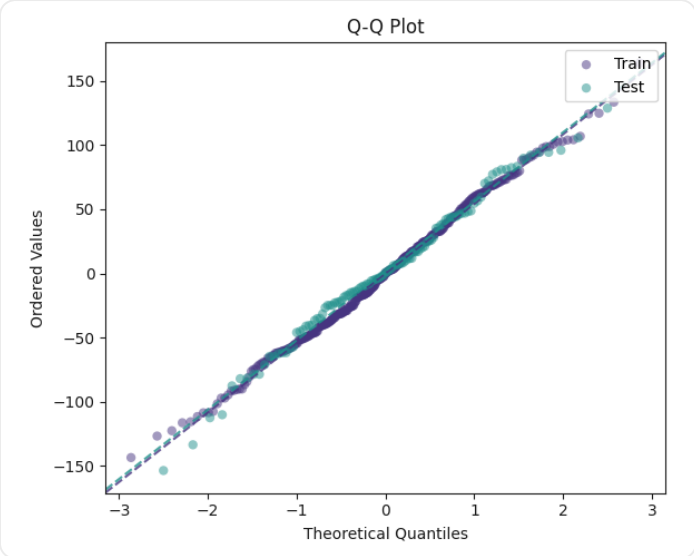
METRICS (TRAIN/TEST SPLIT)

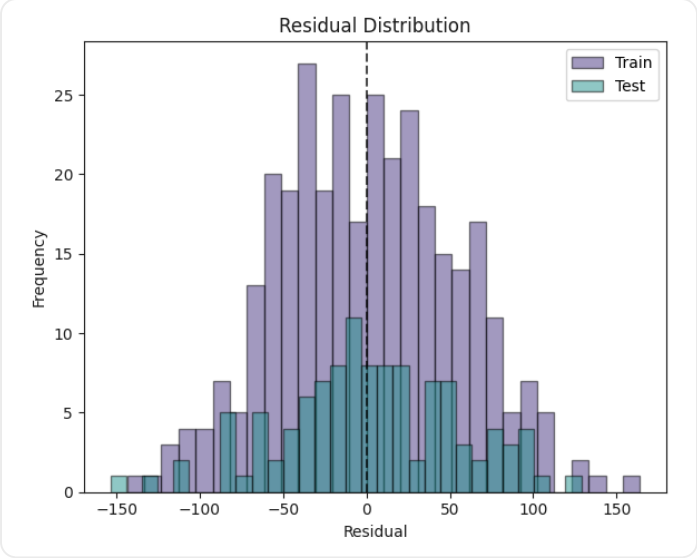
Metric	Train	Test	Gap
Mean Absolute Error	44.0548	41.5485	+2.5063
Max Error	163.9605	153.6572	+10.3033
Median Absolute Error	38.9980	35.2108	+3.7872
Mean Squared Error	2907.2578	2848.3107	+58.9471
R^2 Score	0.5190	0.4849	+0.0341
Root Mean Squared Error	53.9190	53.3696	+0.5494

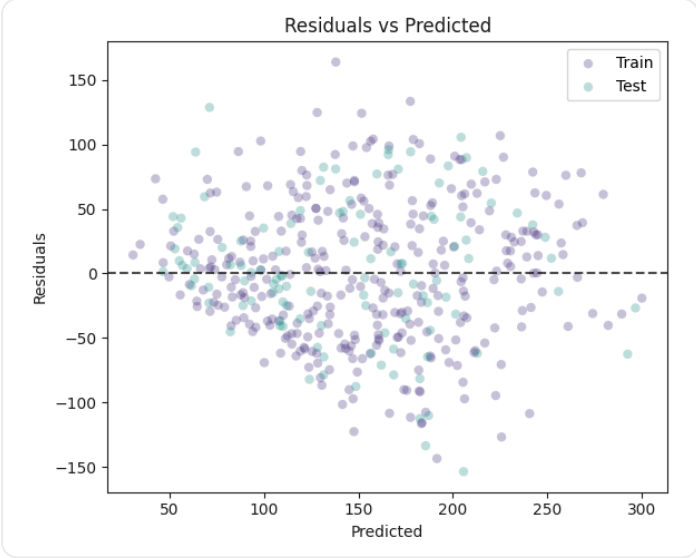
*Gap = Train - Test. Large positive values may indicate overfitting.

VISUALIZATIONS






Residual Distribution
Train
Test
Frequency
Residual
0
5
10
15
20
25
−150
−100
−50
0
50
100
150


Residuals vs Predicted
Train
Test
Residuals
Predicted
150
100
50
0
−50
−100
−150
50
100
150
200
250
300